

Lightest 30.7cm laptop with DVD drive

Panasonic claims its Toughbook S9 to be the world's lightest 30.7cm (12.1-inch) laptop with an integrated DVD drive. Powered by the Intel Core i5-520M vPro processor, the new notebook delivers up to eleven hours of battery life with a standard battery and weighs just 1.36 kg. It also offers optional Gobi2000 mobile broadband and a host of other features designed to maximise return on investment and reduce the total cost of ownership, according to Panasonic.

Despite its wide-screen LED display, the Toughbook S9 survives drops



Panasonic Toughbook S9 laptop

of 30.5cm (12-inch) from any angle using the MIL-STD-810G drop procedure. The spill-resistant keyboard protects against liquid spills of up to 200 cc, and both the lid and base of the notebook can withstand more than 100 kg of pressure.

10,000-rpm hard drive

Hitachi Global Storage Technologies (Hitachi GST) has announced what it claims to be the industry's fastest and most power-efficient enterprise hard drive. The 10,000-rpm, 6Gbps serial-attached SCSI (SAS), 6.4cm enterprise hard drive—the Ultrastar C10K600—

has a capacity of 600 GB and delivers up to 15 per cent better random and 18 per cent faster sequential performance than competing products on the market today, according to the company.

The Ultrastar C10K600 drive is also claimed to set a new standard for energy-efficiency with an operating power specification that is at least 22 per cent lower than the competition. Built on a field-proven platform to ensure greater reliability and reduced qualification time, these compact drives deliver the power, performance and high storage density requirements for Tier 1, enterprise-networked storage arrays, blades and other rack-mounted servers.

The Ultrastar C10K600 drives come in 300GB, 450GB and 600GB versions, providing capacity parity with

8.9cm drives.

Chip to increase solar cell efficiency

STMicroelectronics has announced an innovation for solar panels that allows more of the energy from each cell to reach the power grid. The innovation replaces the simple bypass diode with an intelligent device that enhances efficiency and offers the same package outline. The new device can deliver a fast return-on-investment, in terms of releasing additional energy for solar panel applications.



STMicroelectronics SPV1001 chip for solar panels

The SPV1001 contains a low-loss power switch and a precision controller. Directly replacing bypass diodes, which are used to prevent hotspot effects, it protects the energy normally lost in each diode. Compared to the use of diodes, the integrated power switch provides negligible leakage current when the photovoltaic panel is producing energy. ST's BCD6 chip fabrication process provides the key to this advance by integrating highly efficient power components with logic control circuitry.

Thinnest EMI filters

Texas Instruments has introduced six- and four-channel electromagnetic interference (EMI) filters for LCD applications, the TPD6F202 and TPD4F202. Electrostatic discharge (ESD) ratings exceed the IEC 61000-4-2 specification by at least three times for contact discharge. And both are 67 per cent better than the specification for air-gap discharge.

This allows the signal to be transmitted through the device while it filters out the higher-frequency RF and electromagnetic interferences (RF/EMI) that can appear as lines on a screen, especially LCD.

The devices come in 300µm packages that are less than half the thickness of a credit card, allowing designers to take advantage of previously wasted space under connectors.

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